

Enterprise Data Model Diagram

Grain, Relationships & Filter Direction

This page documents the **semantic data model** for the AI-Augmented Executive Analytics solution.

The model is designed to support:

- executive KPI reporting
- budget vs actual analysis
- product and regional performance analysis
- customer segmentation
- cohort and retention analysis

The objective is to balance **clarity, analytical flexibility, and metric trust**.

1. Model Overview

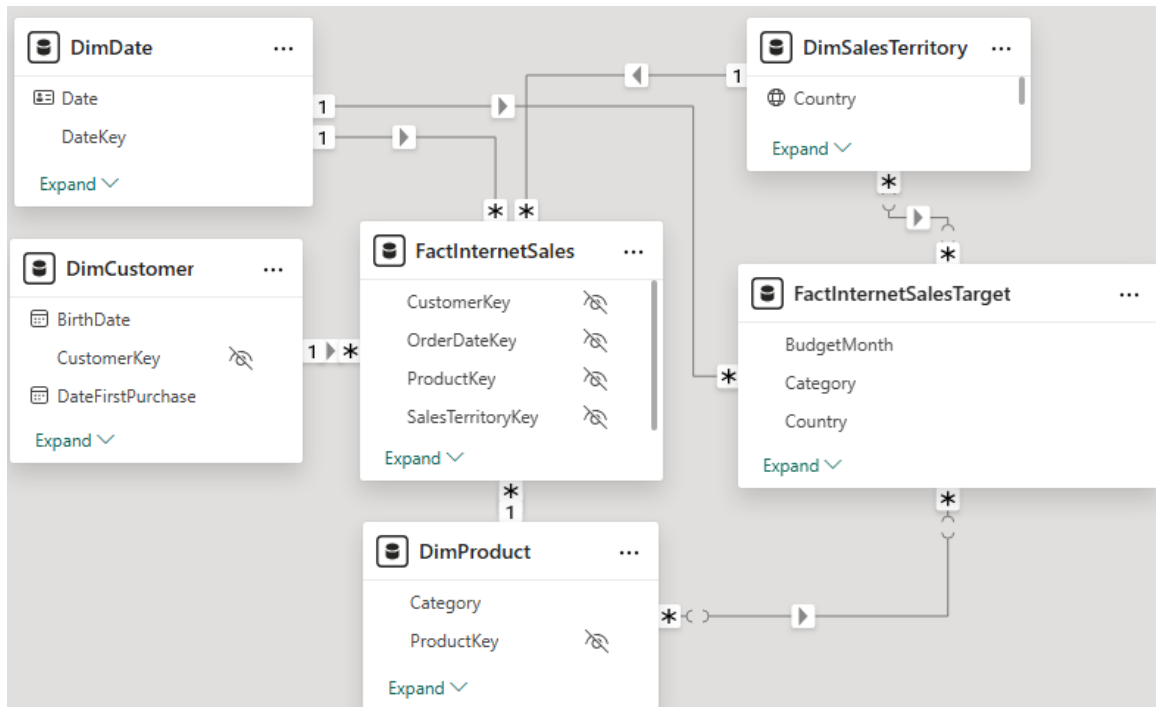
Model Type: Star Schema with controlled many-to-many budget mapping

Storage Mode: Import

Design Goal: Executive-ready analytics with governed Actuals vs Target comparisons

The model contains:

- **2 fact tables**
 - FactInternetSales
 - FactInternetSalesTarget
- **4 core dimensions**
 - DimDate
 - DimProduct
 - DimCustomer
 - DimSalesTerritory



2. Fact Tables & Grain

Fact: `FactInternetSales`

Business Purpose:

Stores actual sales transactions used for revenue, profit, quantity, pricing, customer, and territory analysis.

Grain

One row per sales transaction line

Key Fields

- `OrderDateKey`
- `SalesTerritoryKey`
- `CustomerKey`
- **Product-level sales reference**
- **Transaction metrics:**
 - `SalesAmount`
 - `OrderQuantity`
 - `UnitPrice`
 - `ProductStandardCost`

Why this grain was chosen

- Supports detailed revenue and margin analysis
 - Enables slicing by date, product, customer, and territory
 - Preserves transaction-level fidelity for KPI calculation
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Fact: `FactInternetSalesTarget`

Business Purpose:

Stores budget / target values used for comparing planned performance against actual sales.

Grain

One row per Country × Category × BudgetMonth

Key Fields

- Country
- Category
- BudgetMonth
- budget / target measures

Why this grain was chosen

- Reflects realistic business planning behavior
- Budgets are usually set at summarized levels, not transaction level
- Supports target tracking by geography and product group

Important Modeling Note

This fact table does **not** exist at the same grain as `FactInternetSales`. It is intentionally more aggregated.

That means:

- Actuals can be analyzed at product, subcategory, region, and customer level
 - Targets are only valid at **Country × Category × Month** level
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3. Dimension Tables

Dimension: `DimDate`

Business Purpose:

Provides time intelligence for both actuals and targets.

Grain

One row per calendar date

Attributes

- Date
- Year
- Month
- MonthNo

Model Requirement

- Must be **marked as the Date Table** in Power BI

Why this matters

- Enables DAX time intelligence
 - Standardizes MoM, YoY, rolling trends, and cohort month calculations
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Dimension: DimProduct**Business Purpose:**

Provides product hierarchy for product-level and category-level analysis.

Grain

One row per product

Attributes

- ProductKey
- ProductCategoryKey
- ProductSubcategoryKey
- Category
- Subcategory
- ListPrice

Role in the model

- Supports detailed sales analysis at product level
- Provides category grouping for target comparisons

Important Modeling Note

Because targets are stored at **Category** level rather than `ProductKey`, the connection from product to target is not a simple one-to-many transaction relationship. It is a summarized mapping.

Dimension: `DimCustomer`

Business Purpose:

Supports customer-level analytics, segmentation, and cohort analysis.

Grain

One row per customer

Attributes

- `CustomerKey`
- `Customer Name`
- `Gender`
- `Birth Date`
- `First Purchase Date`

Role in the model

- customer value analysis
 - customer segmentation
 - retention and loyalty analytics
 - cohort assignment logic
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Dimension: `DimSalesTerritory`

Business Purpose:

Provides geographic hierarchy for sales and target reporting.

Grain

One row per sales territory

Attributes

- `SalesTerritoryKey`
- `Region`

- Country
- Group

Role in the model

- regional sales analysis
- country-level target comparison
- geographic drill-down

Important Modeling Note

Targets are stored at **Country** level, while actuals are linked through `SalesTerritoryKey`. This introduces an aggregated mapping between actual sales territories and target countries.

4. Relationship Design

Relationship Summary

From Table	To Table	Join Basis	Cardinality	Active	Direction
FactInternetSales	DimDate	OrderDateKey	Many-to-One	Yes	Single
FactInternetSales	DimCustomer	CustomerKey	Many-to-One	Yes	Single
FactInternetSales	DimSalesTerritory	SalesTerritoryKey	Many-to-One	Yes	Single
FactInternetSales	DimProduct	ProductKey	Many-to-One	Yes	Single
FactInternetSalesTarget	DimDate	BudgetMonth → Date/Month mapping	Many-to-One	Yes	Single
FactInternetSalesTarget	DimProduct	Category mapping	Many-to-Many	Yes	Single / controlled
FactInternetSalesTarget	DimSalesTerritory	Country mapping	Many-to-Many	Yes	Single / controlled

5. Grain Alignment

Actuals Grain

`FactInternetSales` supports:

- Product
- Subcategory
- Customer
- Territory
- Region
- Country
- Day / Month / Year

Target Grain

`FactInternetSalesTarget` supports only:

- Category
- Country
- BudgetMonth

Governance Rule

Budget vs Actual is only valid at the shared grain:

Category × Country × Month

This is the most important rule in the model.

If targets are shown at:

- Product level
- Subcategory level
- Customer level
- Territory below country
- Transaction level

...the results may look precise, but they are not valid.

6. Filter Direction Strategy

Core Principle

All relationships should remain **single-direction** wherever possible.

Why

- avoids ambiguous filter paths
- improves performance
- makes DAX easier to reason about
- reduces risk of duplicated target values

Special Note on Many-to-Many

Because `FactInternetSalesTarget` maps to:

- `DimProduct` through `Category`
- `DimSalesTerritory` through `Country`

...the many-to-many design must be handled carefully.

Recommended Governance Approach

- Use single-direction filters from dimensions to facts
- Restrict target measures with grain-aware DAX
- Prevent invalid target breakdowns below `Category` or `Country`
- Avoid relying on unconstrained many-to-many propagation for executive measures

7. Many-to-Many Design Implications

This model includes **controlled many-to-many relationships** for the target fact.

Why they exist

- Budget data is summarized by category, not product
- Budget data is summarized by country, not territory key

Risks introduced

- duplicated targets when slicing below valid grain
- ambiguous filtering
- misleading variance calculations
- AI narratives making unsupported claims

Required Safeguards

- measures must remove invalid lower-level filters where necessary
 - visuals must avoid unsupported breakdowns
 - documentation must clearly define valid planning grain
 - AI summaries must only reference valid target slices
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8. Date Handling Strategy

`DimDate` is the canonical time table for the model.

Rules

- Mark `DimDate` as the official Date Table
- Use it for all time intelligence calculations
- Map `OrderDateKey` for actuals
- Map `BudgetMonth` for targets at monthly granularity

Why this matters

- consistent MoM / YoY logic
 - stable trend analysis
 - proper cohort month calculations
 - reduced time intelligence errors
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9. Model Governance Rules

1. All KPI measures must clearly state whether they use:
 - Actuals only
 - Targets only
 - Shared-grain variance
 2. Budget measures must only be trusted at:
 - Category
 - Country
 - Month
 - Total
 3. No target-based visual should allow drill-down below valid planning grain.
 4. Many-to-many target relationships must be supported by grain-aware DAX, not assumed to behave safely by default.
 5. AI narrative visuals must not generate commentary for unsupported target slices.
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10. Executive Summary

This model uses a clean star schema for actual sales, with a summarized target table layered on top for planning comparisons.

Actuals are detailed. Targets are aggregated.

The model is governed so executives can compare performance against plan without creating false precision.